
Digital Forensics and Incident Response Report

ImageStegano Challenge Analysis

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Role: SOC Analyst | Incident Response Analyst

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1. Executive Summary

This report presents a comprehensive **Digital Forensics and Incident Response (DFIR)** investigation conducted on a steganography-based challenge from the LetsDefend platform.

The objective was to analyze a suspicious PNG image file, identify any **hidden malicious payloads**, determine the **embedding technique**, and extract actionable **Indicators of Compromise (IoCs)**.

The investigation successfully revealed that the image contained a **hidden PowerShell payload embedded using steganography (LSB technique)**. Further analysis confirmed that the payload was a **heavily obfuscated variant of the Mimikatz credential dumping tool**, commonly used in post-exploitation phases.

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2. Investigation Scope and Initial Triage

Primary Artifact

- **Archive Name:** Im493.zip
- **Extracted File:** Sd6wF1A1v.png
- **File Size:** 64,849 KB

Initial Assessment

The PNG file size was **significantly larger than expected**, which strongly suggested:

- Presence of **hidden embedded data**
- Possible **steganographic manipulation**
- Potential **payload delivery mechanism**

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3. Detailed Investigation Procedure

Phase 1: Metadata Analysis

Tool Used

- exiftool (via WSL environment)

Command

wsl exiftool Sd6wF1A1v.png

Key Findings

- **Device Manufacturer:** Hewlett-Packard
- **CMM Type:** Linotronic

Analysis Insight

Metadata provided indicators about the **origin environment**, which can sometimes help in:

- Attribution
- Tool fingerprinting
- Identifying attacker artifacts

Phase 2: Steganography Technique Identification

Methodology

OSINT-based investigation using targeted queries:

“PowerShell tool to embed payload into image”

Key Finding

- Identified tool: **Invoke-PSImage**

Technical Explanation (مختصر عربي + إنجليزي)

- **Steganography (إخفاء البيانات داخل ملفات):**
  هي تقنية لإخفاء بيانات داخل صورة بدون تغيير واضح في الشكل.

- **LSB Encoding (Least Significant Bit):**
  يتم تخزين البيانات داخل أقل بت في كل pixelمما يجعلها غير ملحوظة، .

👉 Tool behavior:

- Encodes PowerShell script into image pixels
- Executes payload **in-memory (fileless execution)**
- Helps bypass traditional detection systems

Phase 3: Payload Extraction

Tool Used

- Custom Python decoder: psimage_decoder.py

Execution Command

py .\3omdaScript.py .\Sd6wF1A1v.png 3omda.txt

Result

- Successfully extracted:
  - **Obfuscated PowerShell script**
  - Encoded using **Base64**
  - Designed for **in-memory execution**

Phase 4: Malware Analysis & IoCs

Analysis Approach

- Static analysis of decoded script
- Hash lookup via threat intelligence platforms

Key Findings

- **Malware Family:** Mimikatz
- **Detection Rate:** 58 / 72 engines
- **Primary Function:** Credential Dumping

Critical Function Identified

- invoke-mimikatz
- Export reference: powershell_reflective_mimikatz

Executable Details

- **Architecture:** 32-bit
- **SHA-256 Hash:**

be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059

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4. Key Findings Summary

Parameter	Result
Device Manufacturer	Hewlett-Packard
CMM Type	Linotronic
Steganography Tool	Invoke-PSImage
Payload Function	invoke-mimikatz
Malware Family	Mimikatz
SHA-256 Hash	be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059

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5. Conclusion

The investigation confirmed that **Sd6wF1A1v.png** was a **malicious steganographic carrier file** rather than a legitimate image.

Attackers leveraged:

- **Invoke-PSImage** for payload embedding

- **LSB steganography** for concealment
- **PowerShell-based in-memory execution** for evasion

The embedded payload was identified as **Mimikatz**, a well-known **post-exploitation tool used for credential harvesting and lateral movement**.

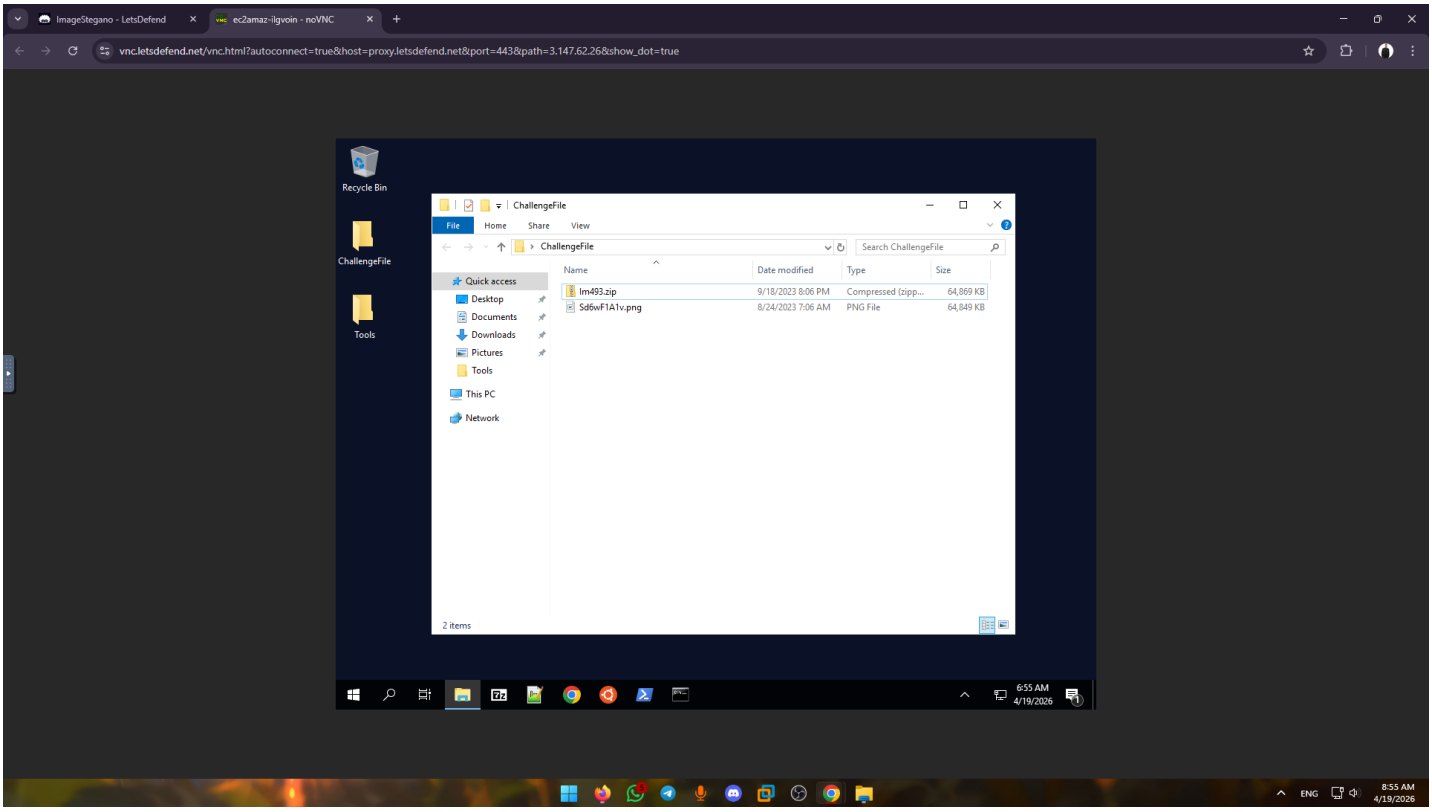
Key Security Insight (مهم جدًا)

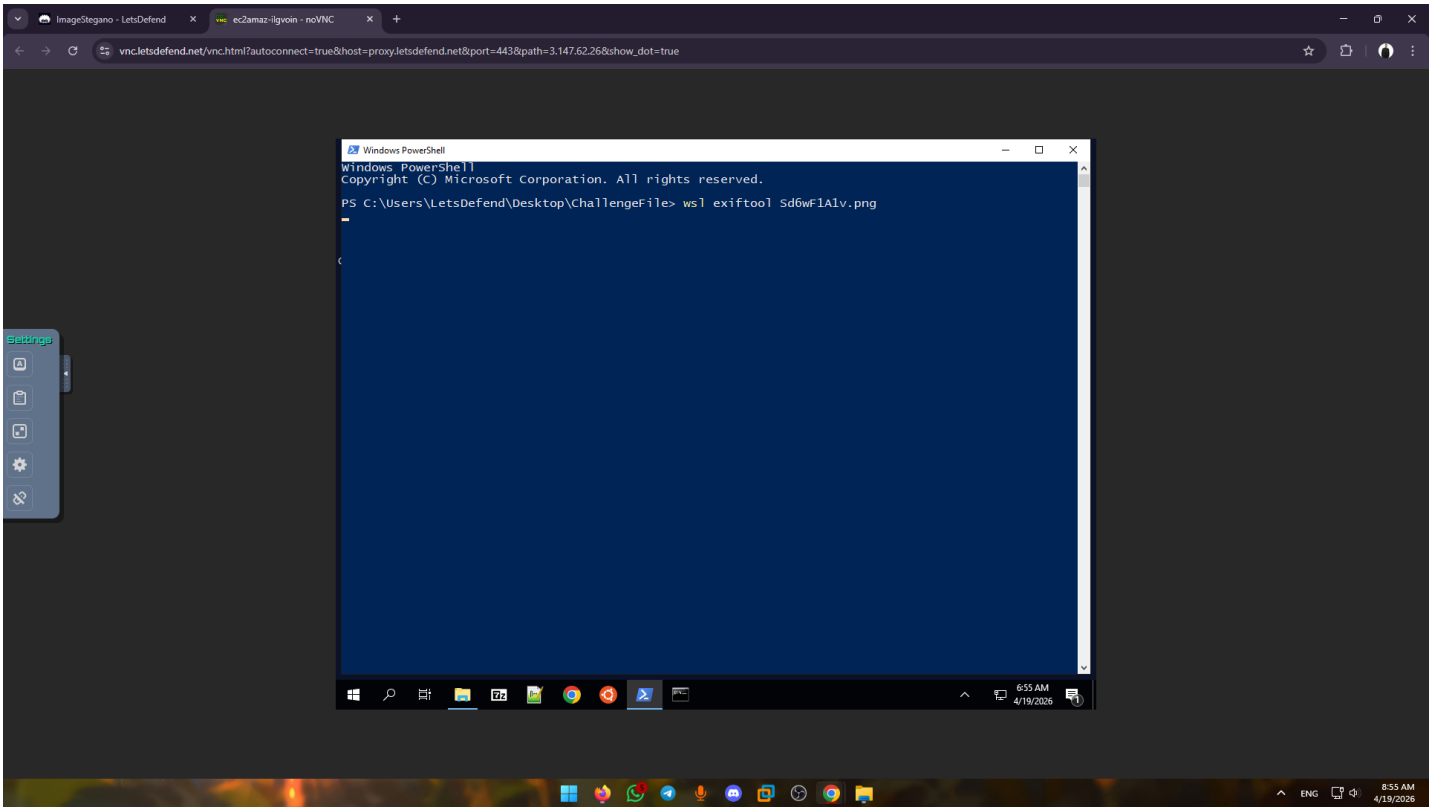
- الملفات الكبيرة غير الطبيعية لازم تتحلل
- آمن JPG أو PNG مش كل
- لازم تفحص:
    - Metadata
    - File size anomalies
    - Hidden payloads (LSB analysis)

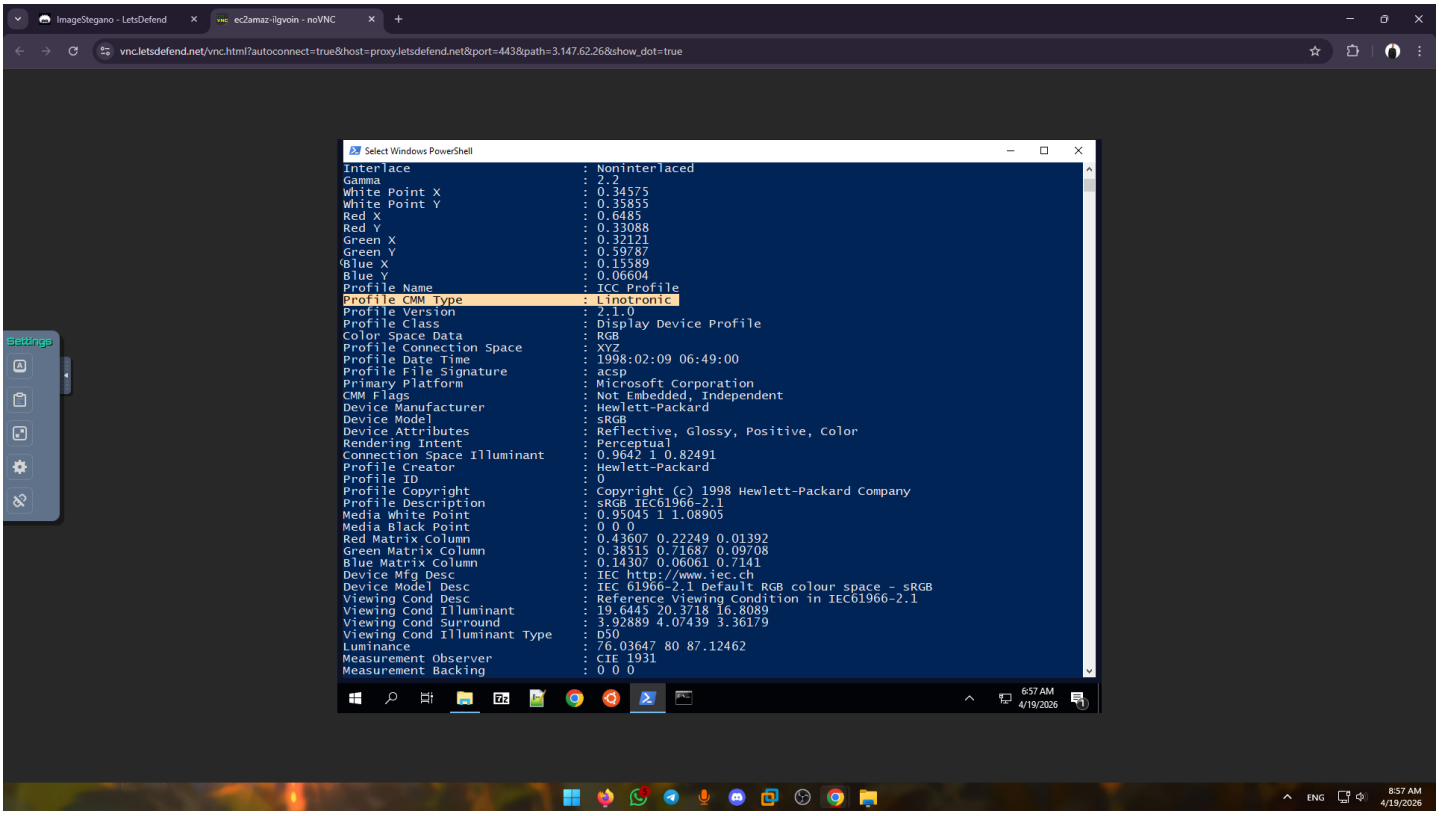
Final Takeaway

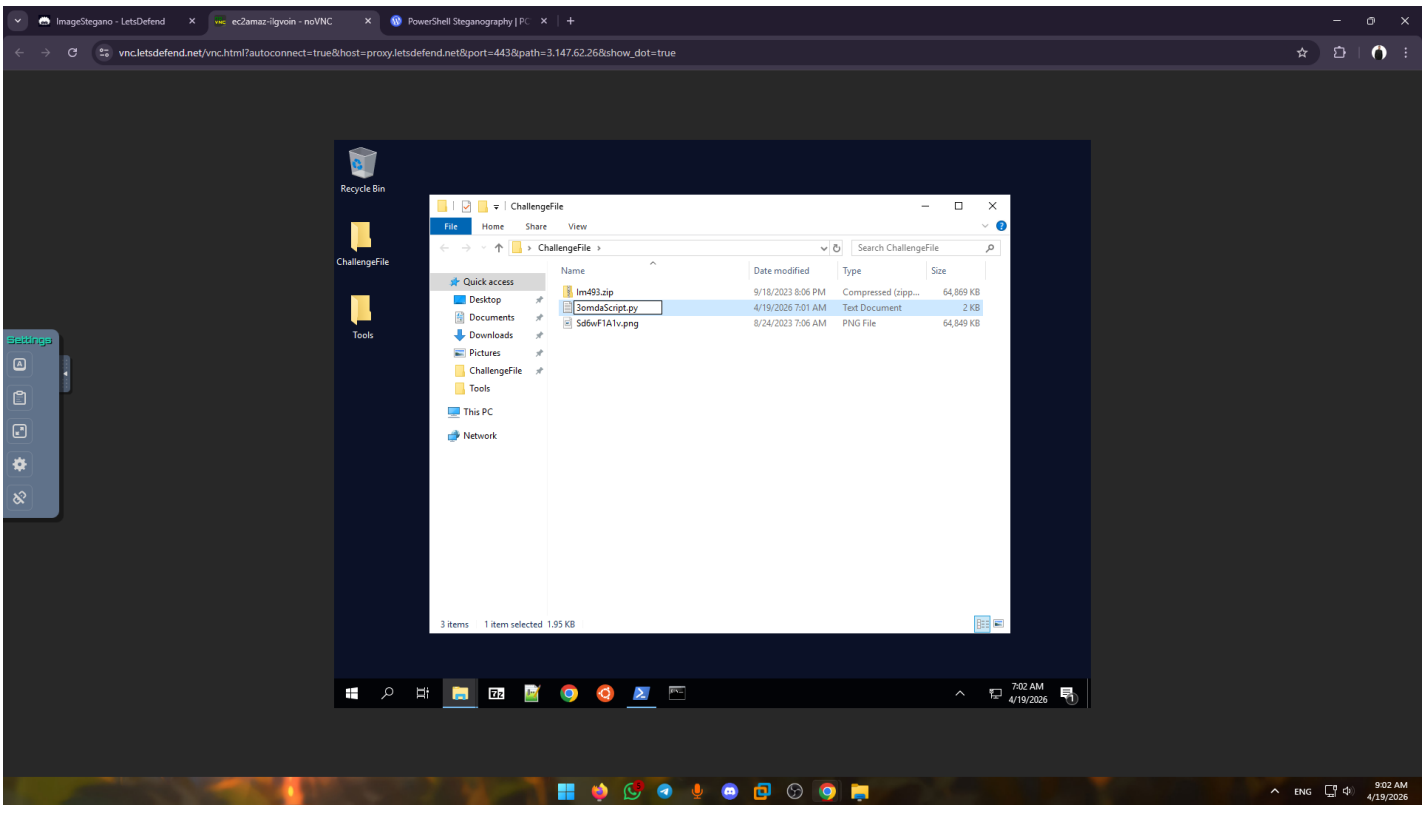
This case highlights the importance of:

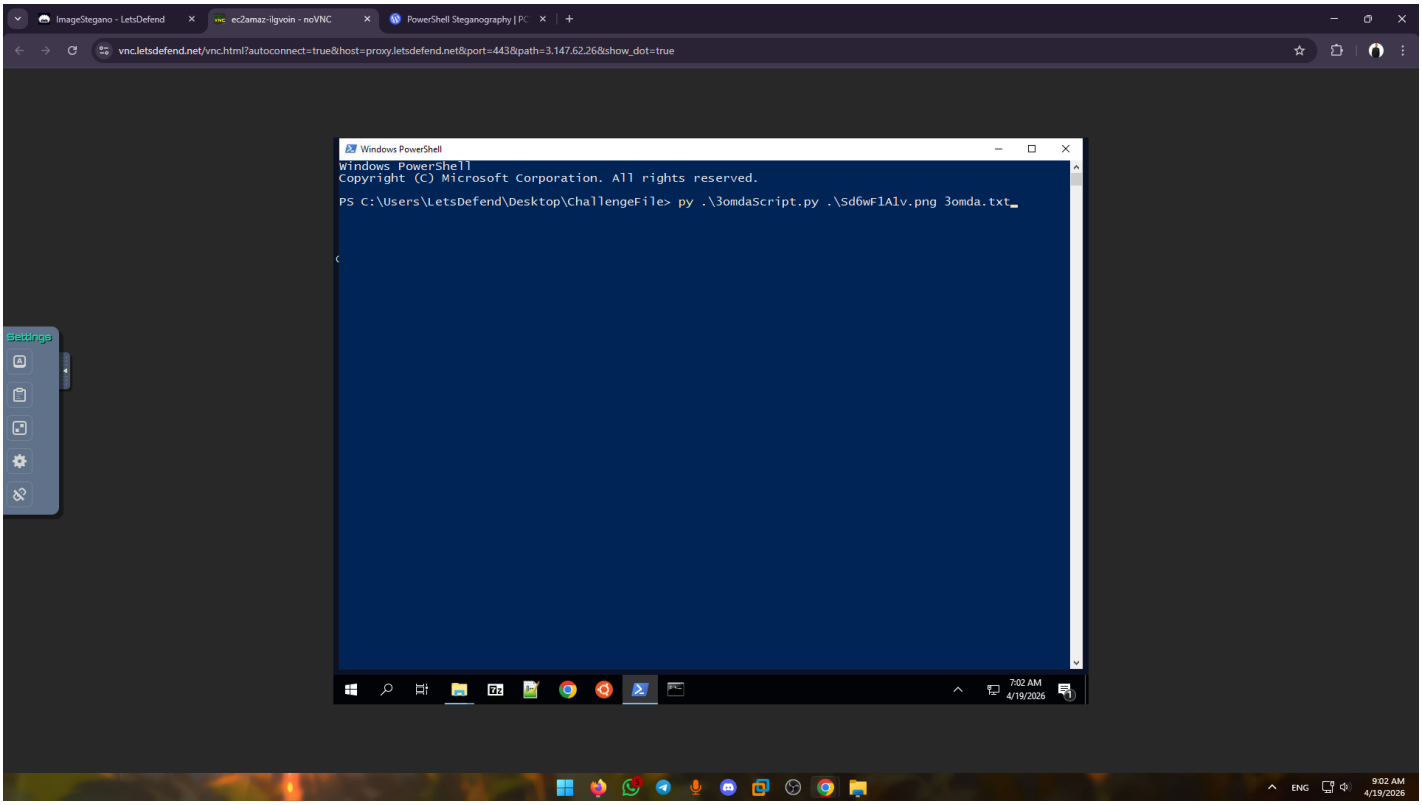
- Deep forensic analysis beyond file extensions
- Detecting **fileless malware techniques**
- Understanding attacker tradecraft in **steganography-based attacks**

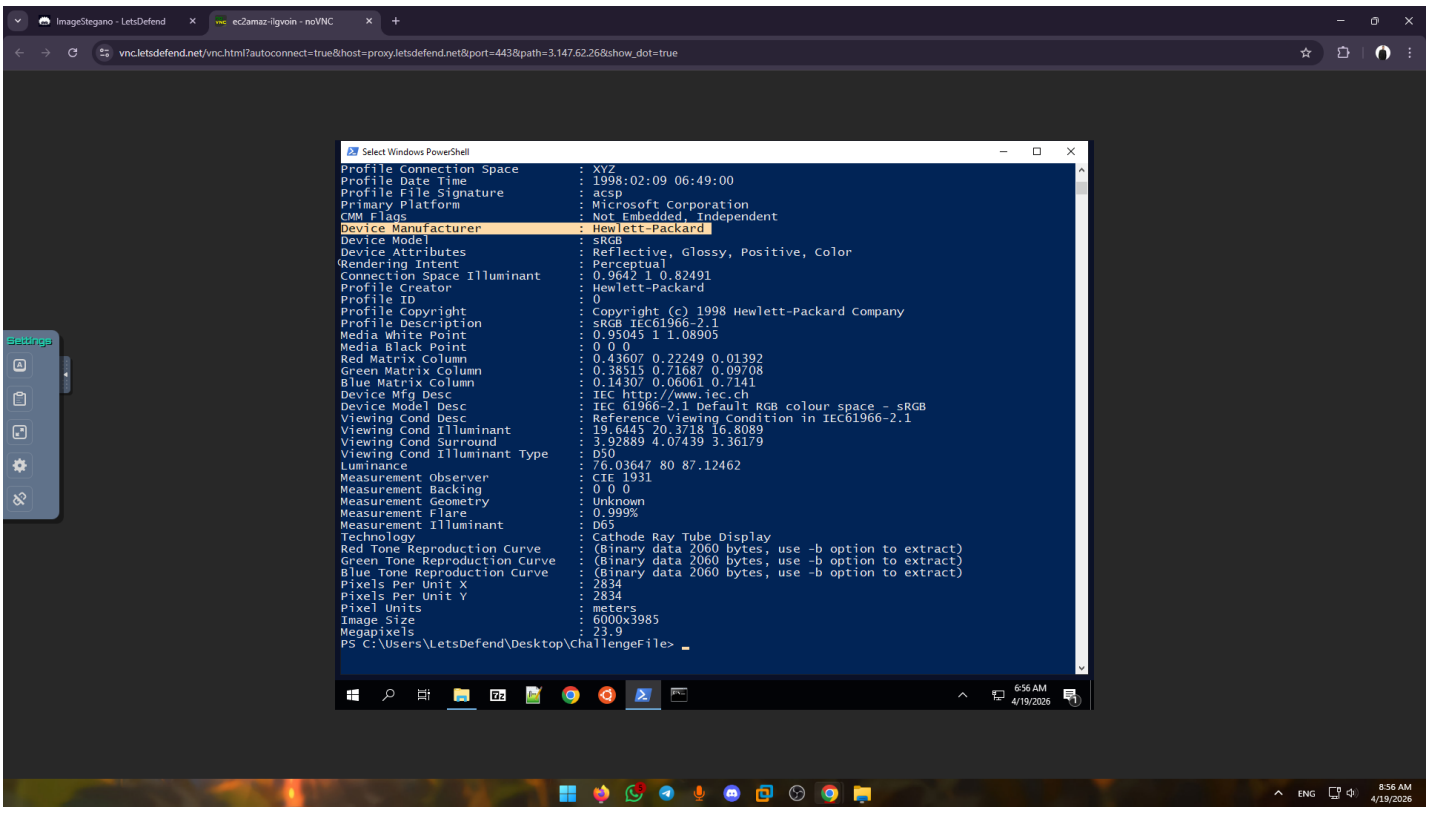

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Recycle Bin
ChallengeFile
Tools
ChallengeFile
File
Home
Share
View
Search ChallengeFile
Quick access
Desktop
Documents
Downloads
Pictures
Tools
This PC
Network
Name
Date modified
Type
Size
Im493.zip
9/18/2023 8:06 PM
Compressed (zipp...
64,869 KB
Sd6wF1A1v.png
8/24/2023 7:06 AM
PNG File
64,849 KB
2 items
6:55 AM
4/19/2026
ENG
8:55 AM
4/19/2026

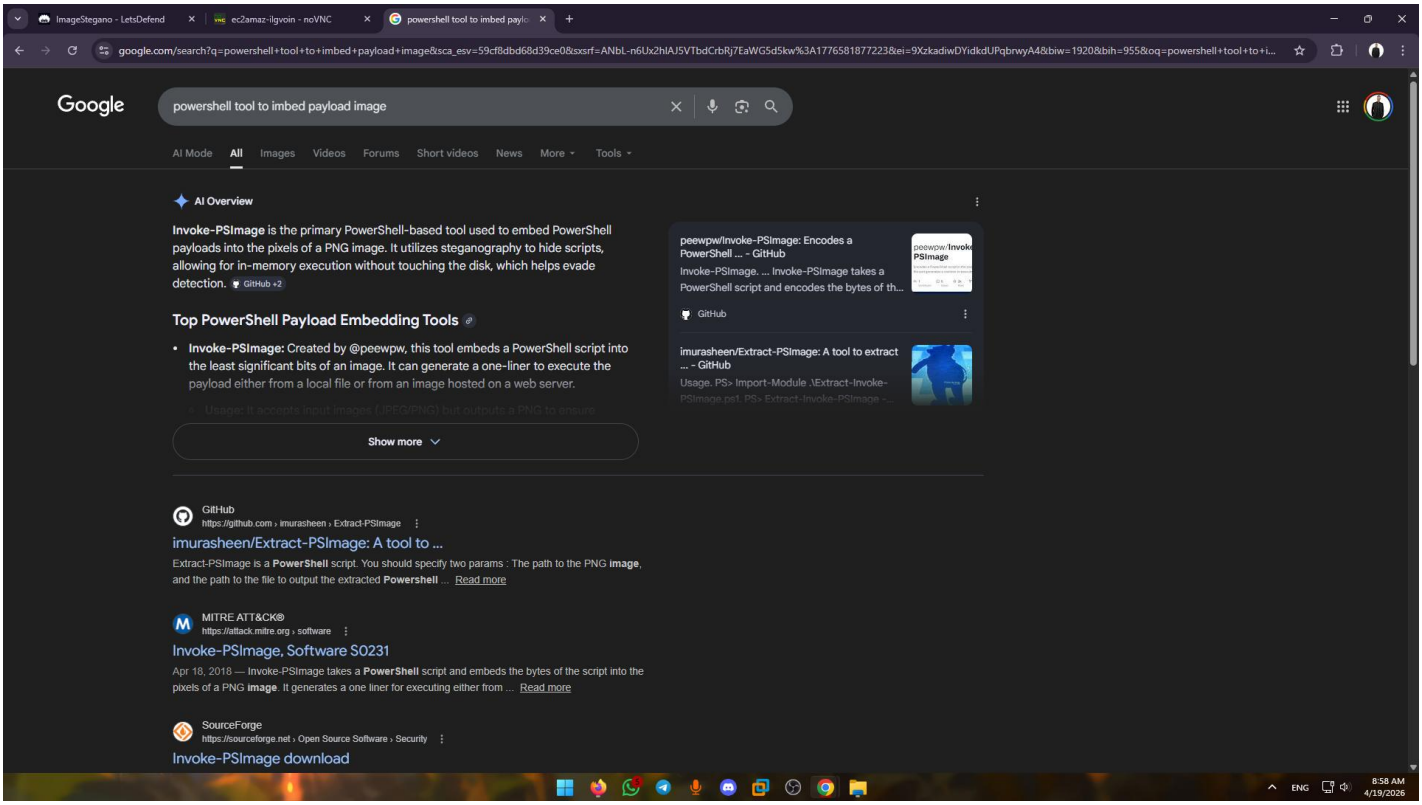

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\LetsDefend\Desktop\ChallengeFile> wsl exiftool Sd6wF1A1v.png
Settings
6:55 AM
4/19/2026
ENG
8:55 AM
4/19/2026

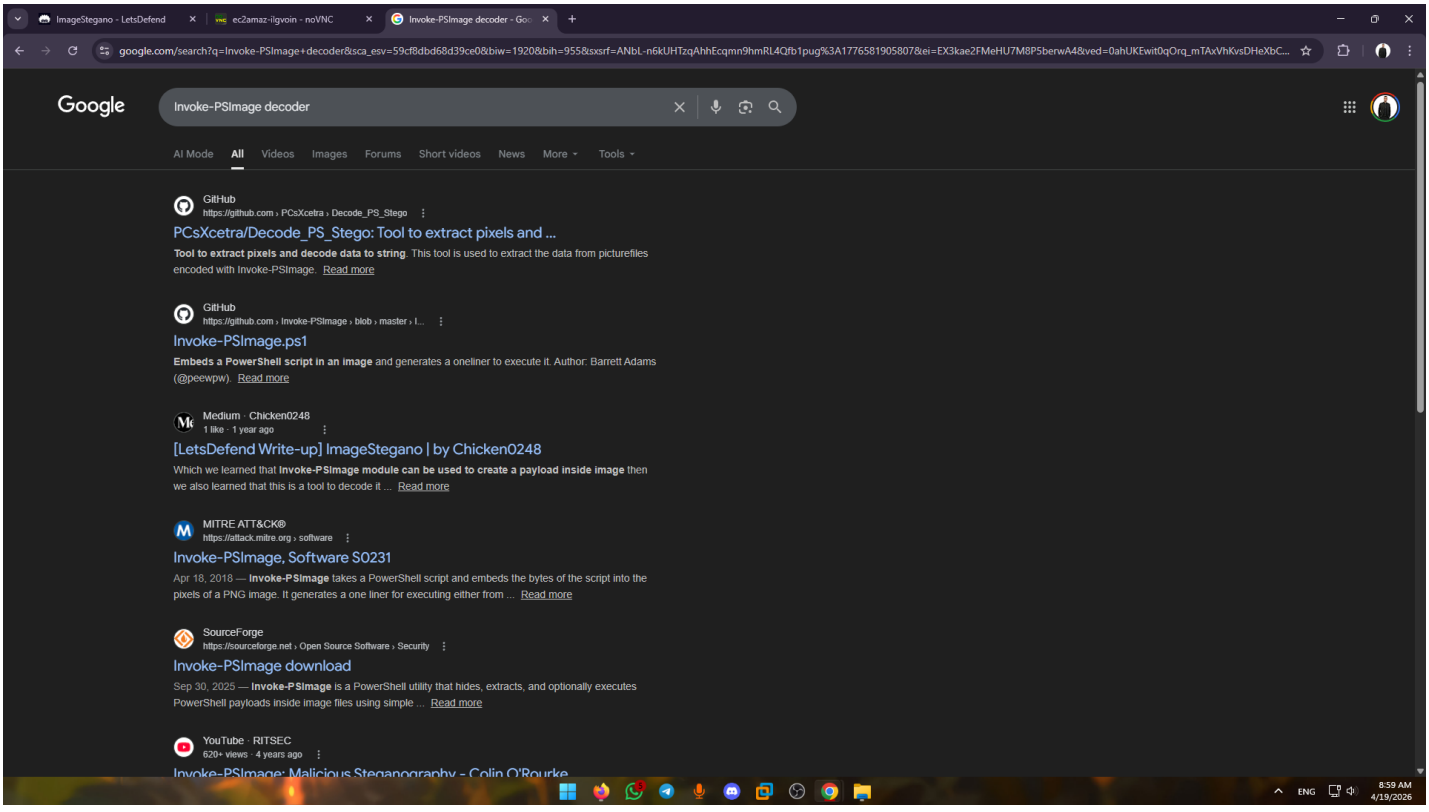

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Select Windows PowerShell
Interlace : Noninterlaced
Gamma : 2.2
White Point X : 0.34575
White Point Y : 0.35855
Red X : 0.6485
Red Y : 0.33088
Green X : 0.32121
Green Y : 0.59787
Blue X : 0.15589
Blue Y : 0.06604
Profile Name : ICC Profile
Profile CMM Type : Linotronic
Profile Version : 2.1.0
Profile Class : Display Device Profile
Color Space Data : RGB
Profile Connection Space : XYZ
Profile Date Time : 1998:02:09 06:49:00
Profile File Signature : acsp
Primary Platform : Microsoft Corporation
CMM Flags : Not Embedded, Independent
Device Manufacturer : Hewlett-Packard
Device Model : sRGB
Device Attributes : Reflective, Glossy, Positive, Color
Rendering Intent : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator : Hewlett-Packard
Profile ID : 0
Profile Copyright : Copyright (c) 1998 Hewlett-Packard Company
Profile Description : sRGB IEC61966-2.1
Media White Point : 0.95045 1 1.08905
Media Black Point : 0 0 0
Red Matrix Column : 0.43607 0.22249 0.01392
Green Matrix Column : 0.38515 0.71687 0.09708
Blue Matrix Column : 0.14307 0.06061 0.7141
Device Mfg Desc : IEC http://www.iec.ch
Device Model Desc : IEC 61966-2.1 Default RGB colour space - sRGB
Viewing Cond Desc : Reference Viewing Condition in IEC61966-2.1
Viewing Cond Illuminant : 19.6445 20.3718 16.8089
Viewing Cond Surround : 3.92889 4.07439 3.36179
Viewing Cond Illuminant Type : D50
Luminance : 76.03647 80 87.12462
Measurement Observer : CIE 1931
Measurement Backing : 0 0 0
Settings
6:57 AM
4/19/2026
ENG
8:57 AM
4/19/2026

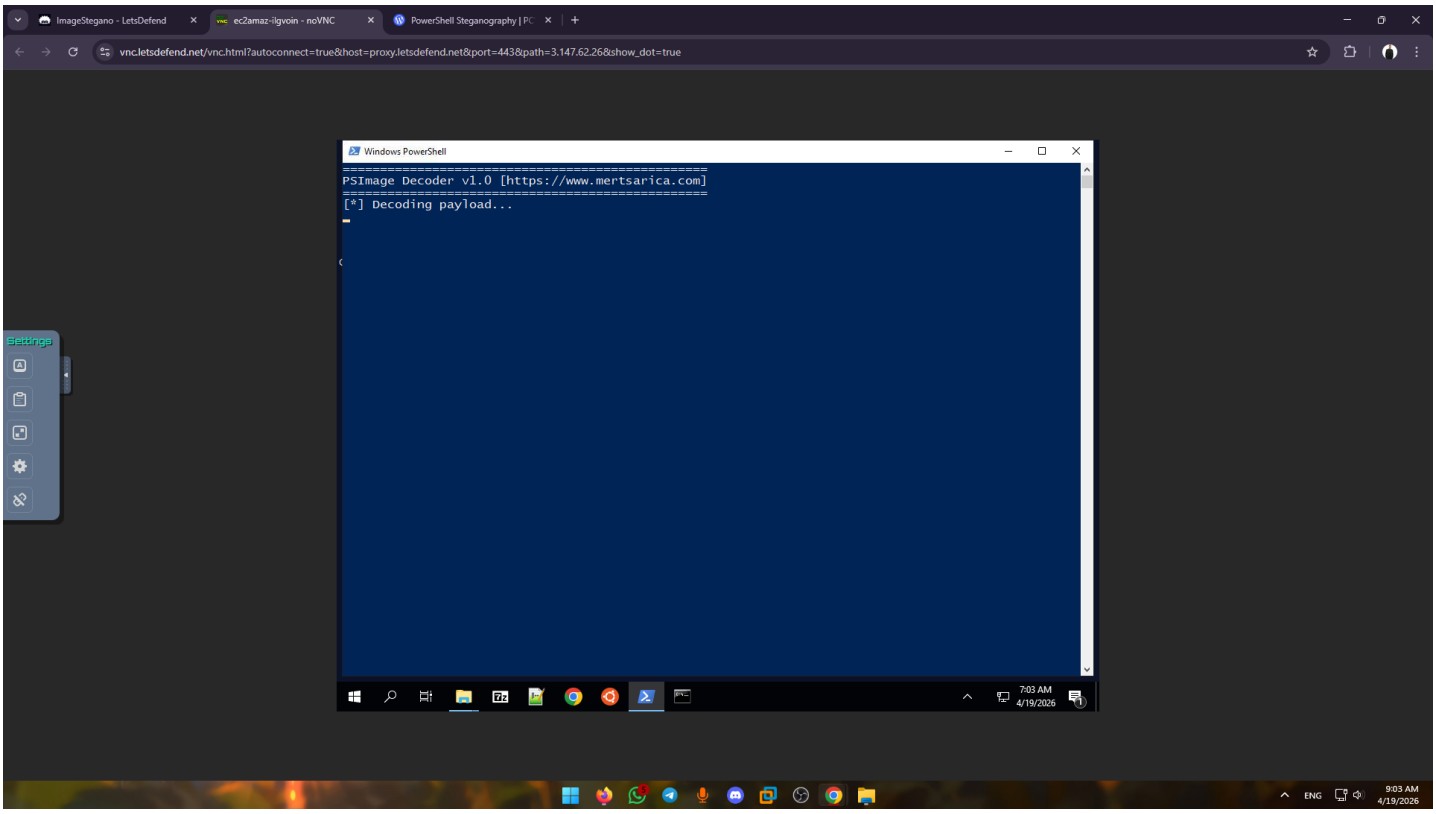

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Recycle Bin
ChallengeFile
Tools
ChallengeFile
File
Home
Share
View
Search ChallengeFile
Quick access
Desktop
Documents
Downloads
Pictures
ChallengeFile
Tools
This PC
Network
Name
Date modified
Type
Size
lm493.zip
9/18/2023 8:06 PM
Compressed (zipp...
64,869 KB
3omdaScript.py
4/19/2026 7:01 AM
Text Document
2 KB
Sd6wF1A1v.png
8/24/2023 7:06 AM
PNG File
64,849 KB
3 items
1 item selected 1.95 KB
Settings
7:02 AM
4/19/2026
ENG
9:02 AM
4/19/2026

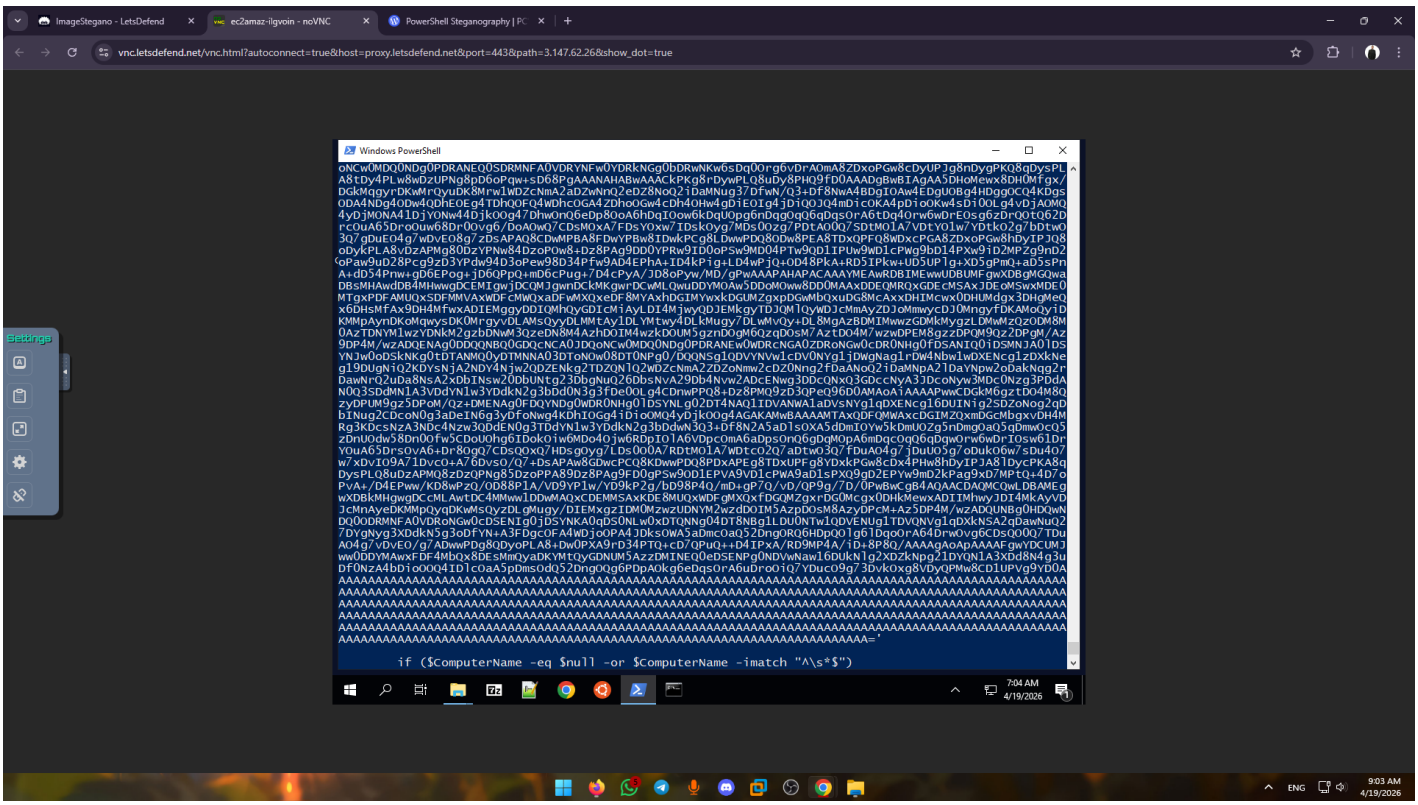

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\LetsDefend\Desktop\ChallengeFile> py .\3omdaScript.py .\Sd6wF1A1v.png 3omda.txt
Settings
7:02 AM
4/19/2026
ENG
9:02 AM
4/19/2026

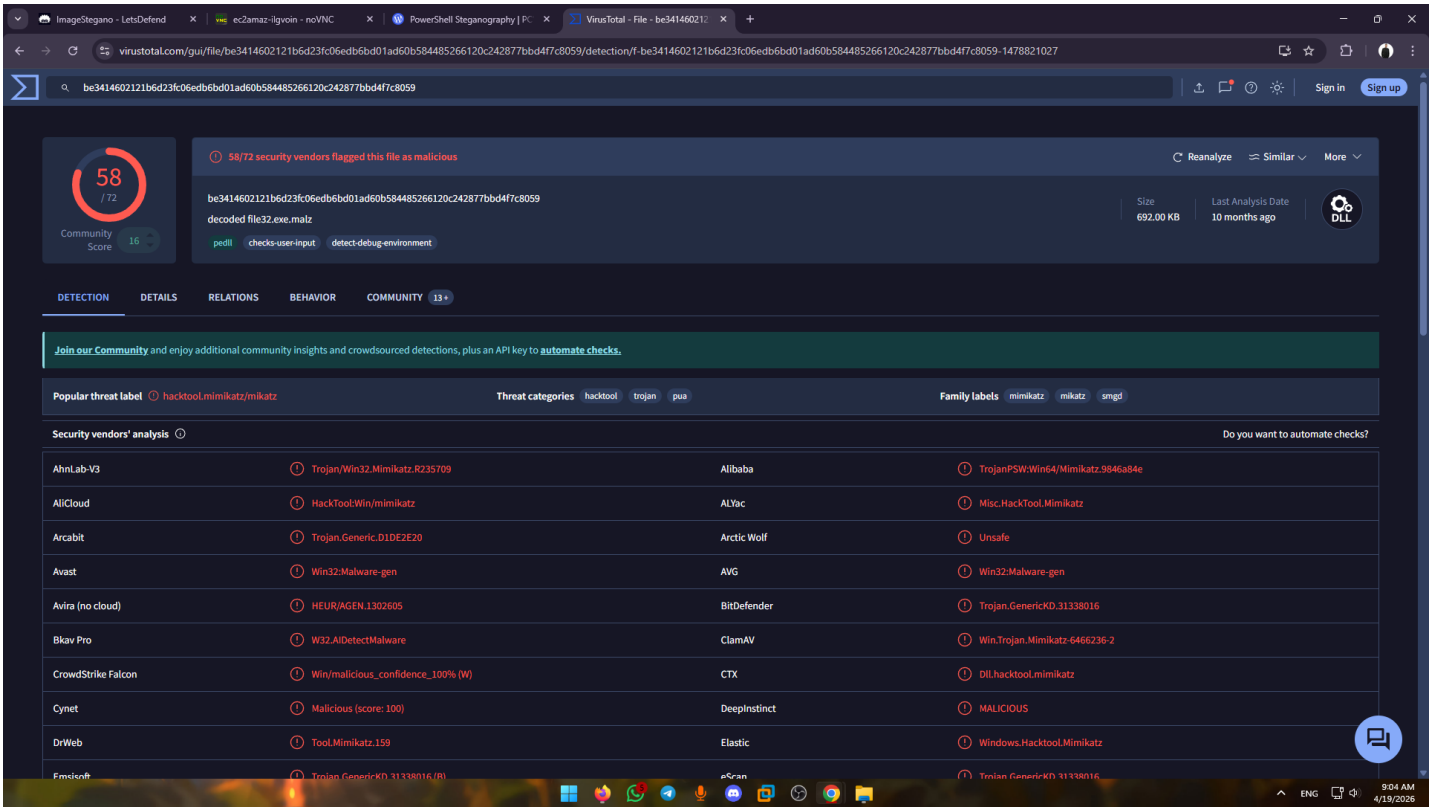

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Select Windows PowerShell
Profile Connection Space : XYZ
Profile Date Time : 1998:02:09 06:49:00
Profile File Signature : acsp
Primary Platform : Microsoft Corporation
CMM Flags : Not Embedded, Independent
Device Manufacturer : Hewlett-Packard
Device Model : sRGB
Device Attributes : Reflective, Glossy, Positive, Color
Rendering Intent : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator : Hewlett-Packard
Profile ID : 0
Profile Copyright : Copyright (c) 1998 Hewlett-Packard Company
Profile Description : sRGB IEC61966-2.1
Media White Point : 0.95045 1 1.08905
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Red Matrix Column : 0.43607 0.22249 0.01392
Green Matrix Column : 0.38515 0.71687 0.09708
Blue Matrix Column : 0.14307 0.06061 0.7141
Device Mfg Desc : IEC http://www.iec.ch
Device Model Desc : IEC 61966-2.1 Default RGB colour space - sRGB
Viewing Cond Desc : Reference Viewing Condition in IEC61966-2.1
Viewing Cond Illuminant : 19.6445 20.3718 16.8089
Viewing Cond Surround : 3.92889 4.07439 3.36179
Viewing Cond Illuminant Type : D50
Luminance : 76.03647 80 87.12462
Measurement Observer : CIE 1931
Measurement Backing : 0 0 0
Measurement Geometry : Unknown
Measurement Flare : 0.999%
Measurement Illuminant : D65
Technology : Cathode Ray Tube Display
Red Tone Reproduction Curve : (Binary data 2060 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 2060 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 2060 bytes, use -b option to extract)
Pixels Per Unit X : 2834
Pixels Per Unit Y : 2834
Pixel Units : meters
Image Size : 6000x3985
Megapixels : 23.9
PS C:\Users\LetsDefend\Desktop\ChallengeFile>
Settings
6:56 AM
4/19/2026
ENG
8:56 AM
4/19/2026

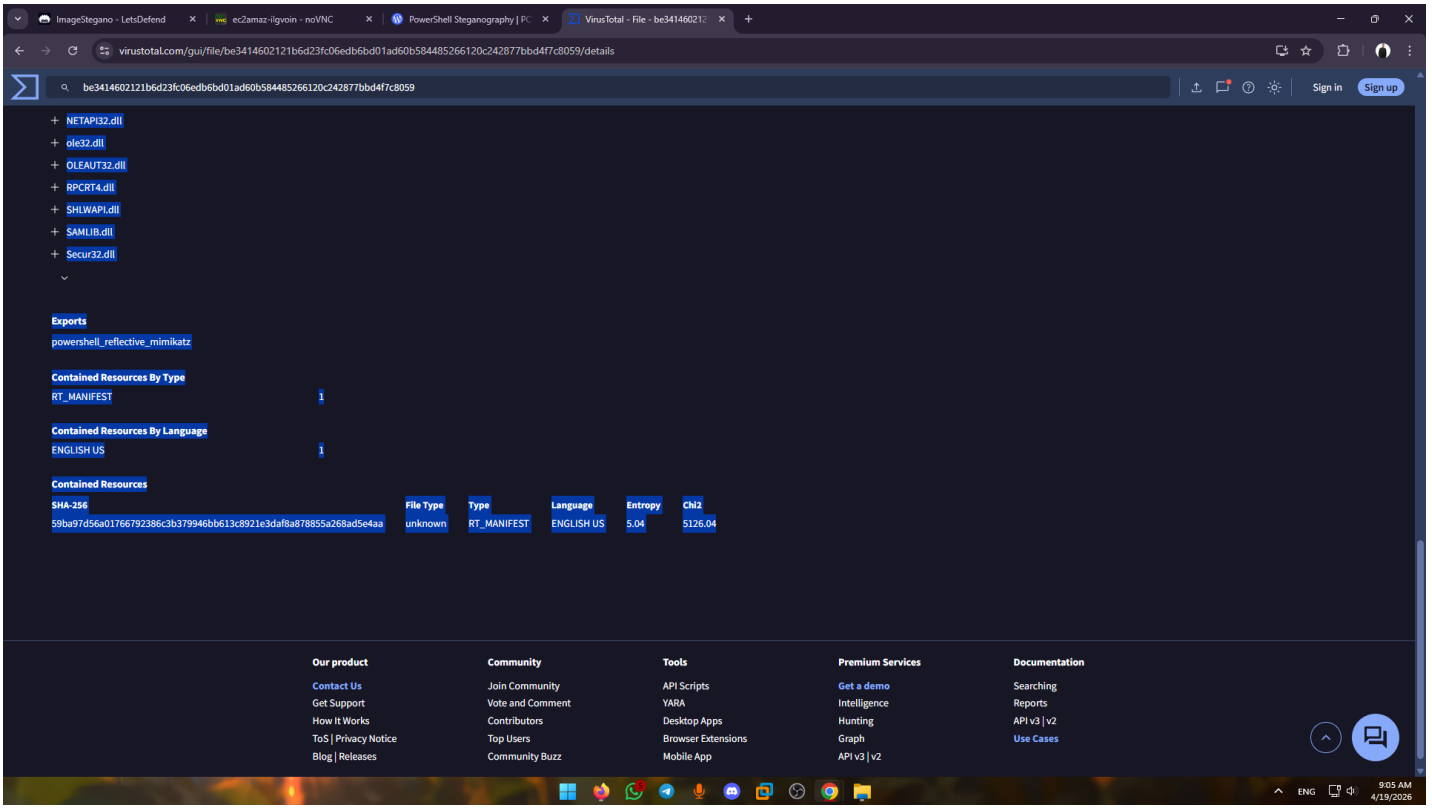

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
powershell tool to imbed payl
google.com/search?q=powershell+tool+to+imbed+payload+image&sca_esv=59cf8dbd68d39ce0&sxsrf=ANbL-n6Ux2hlAJ5VTbdCrbRj7EaWG5d5kw%3A1776581877223&ei=9XzkadiwDYidkdUPqbrwyA4&biw=1920&bih=955&oq=powershell+tool+to+i...
Google
powershell tool to imbed payload image
AI Mode
All
Images
Videos
Forums
Short videos
News
More
Tools
AI Overview
Invoke-PSImage is the primary PowerShell-based tool used to embed PowerShell payloads into the pixels of a PNG image. It utilizes steganography to hide scripts, allowing for in-memory execution without touching the disk, which helps evade detection.
GitHub +2
Top PowerShell Payload Embedding Tools
Invoke-PSImage: Created by @peewpw, this tool embeds a PowerShell script into the least significant bits of an image. It can generate a one-liner to execute the payload either from a local file or from an image hosted on a web server.
peewpw/Invoke-PSImage: Encodes a PowerShell ... - GitHub
Invoke-PSImage. ... Invoke-PSImage takes a PowerShell script and encodes the bytes of th...
GitHub
imurasheen/Extract-PSImage: A tool to extract ... - GitHub
Usage. PS> Import-Module .\Extract-Invoke-PSImage.psd1. PS> Extract-Invoke-PSImage -...
Show more
GitHub
https://github.com › imurasheen › Extract-PSImage
imurasheen/Extract-PSImage: A tool to ...
Extract-PSImage is a PowerShell script. You should specify two params : The path to the PNG image, and the path to the file to output the extracted Powershell ... Read more
MITRE ATT&CK®
https://attack.mitre.org › software
Invoke-PSImage, Software S0231
Apr 18, 2018 — Invoke-PSImage takes a PowerShell script and embeds the bytes of the script into the pixels of a PNG image. It generates a one liner for executing either from ... Read more
SourceForge
https://sourceforge.net › Open Source Software › Security
Invoke-PSImage download
ENG
8:58 AM
4/19/2026

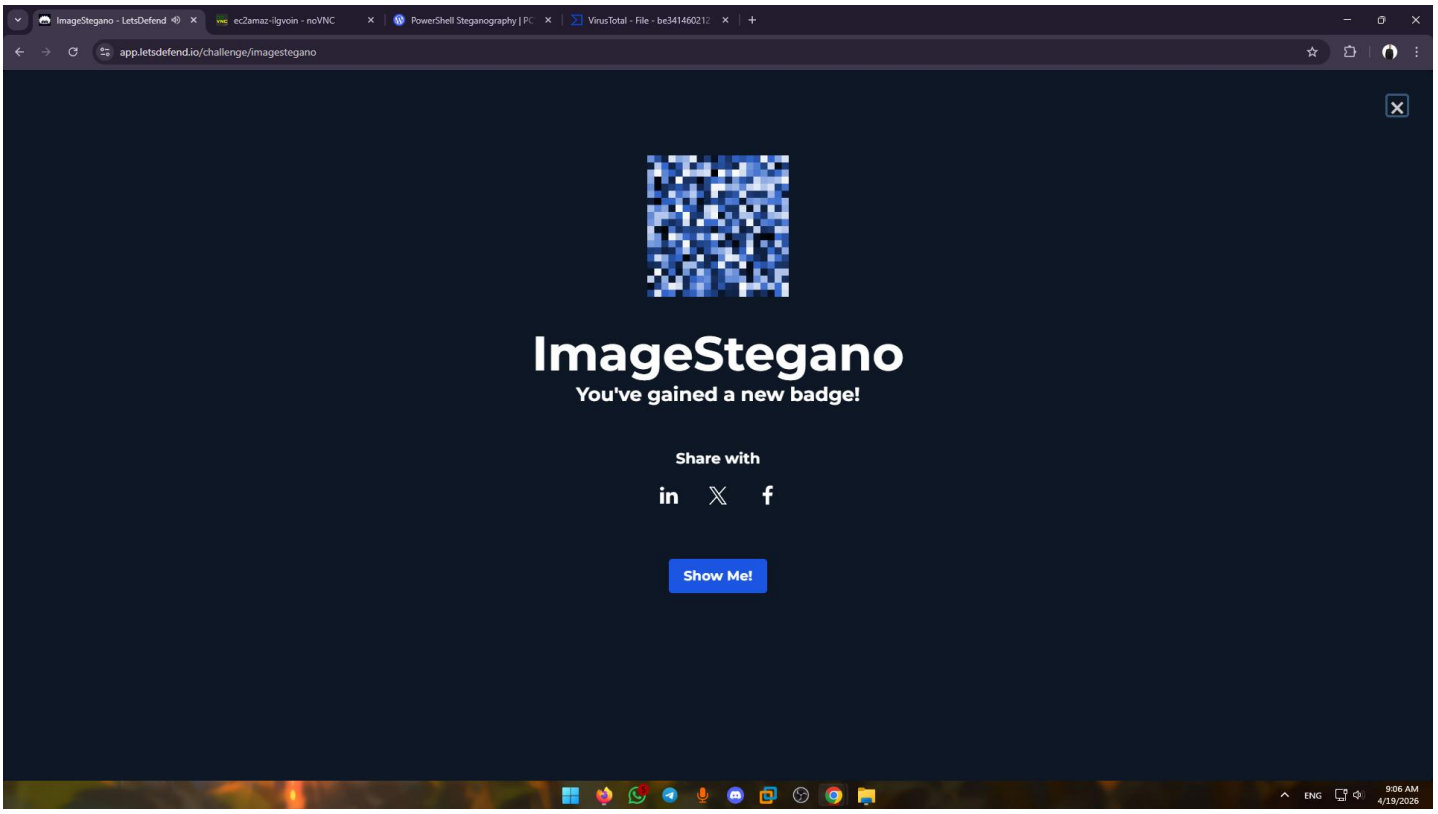

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
Invoke-PSImage decoder - Goo
google.com/search?q=Invoke-PSImage+decoder&sca_esv=59cf8dbd68d39ce0&biw=1920&bih=955&sxsrf=ANbL-n6kUHTzqAhhEcqmn9hmRL4Qfb1pug%3A1776581905807&ei=EX3kae2FMeHU7M8P5berwA4&ved=0ahUKEwit0qOrq_mTAxVhKvsDHeXbC...
Google
Invoke-PSImage decoder
AI Mode
All
Videos
Images
Forums
Short videos
News
More
Tools
GitHub
https://github.com › PCsXcetra › Decode_PS_Stego
PCsXcetra/Decode_PS_Stego: Tool to extract pixels and ...
Tool to extract pixels and decode data to string. This tool is used to extract the data from picturefiles encoded with Invoke-PSImage. Read more
GitHub
https://github.com › Invoke-PSImage › blob › master › I...
Invoke-PSImage.ps1
Embeds a PowerShell script in an image and generates a oneliner to execute it. Author: Barrett Adams (@peewpw). Read more
Medium · Chicken0248
1 like · 1 year ago
[LetsDefend Write-up] ImageStegano | by Chicken0248
Which we learned that Invoke-PSImage module can be used to create a payload inside image then we also learned that this is a tool to decode it ... Read more
MITRE ATT&CK®
https://attack.mitre.org › software
Invoke-PSImage, Software S0231
Apr 18, 2018 — Invoke-PSImage takes a PowerShell script and embeds the bytes of the script into the pixels of a PNG image. It generates a one liner for executing either from ... Read more
SourceForge
https://sourceforge.net › Open Source Software › Security
Invoke-PSImage download
Sep 30, 2025 — Invoke-PSImage is a PowerShell utility that hides, extracts, and optionally executes PowerShell payloads inside image files using simple ... Read more
YouTube · RITSEC
620+ views · 4 years ago
Invoke-PSImage: Malicious Steganography - Colin O'Rourke
ENG
8:59 AM
4/19/2026

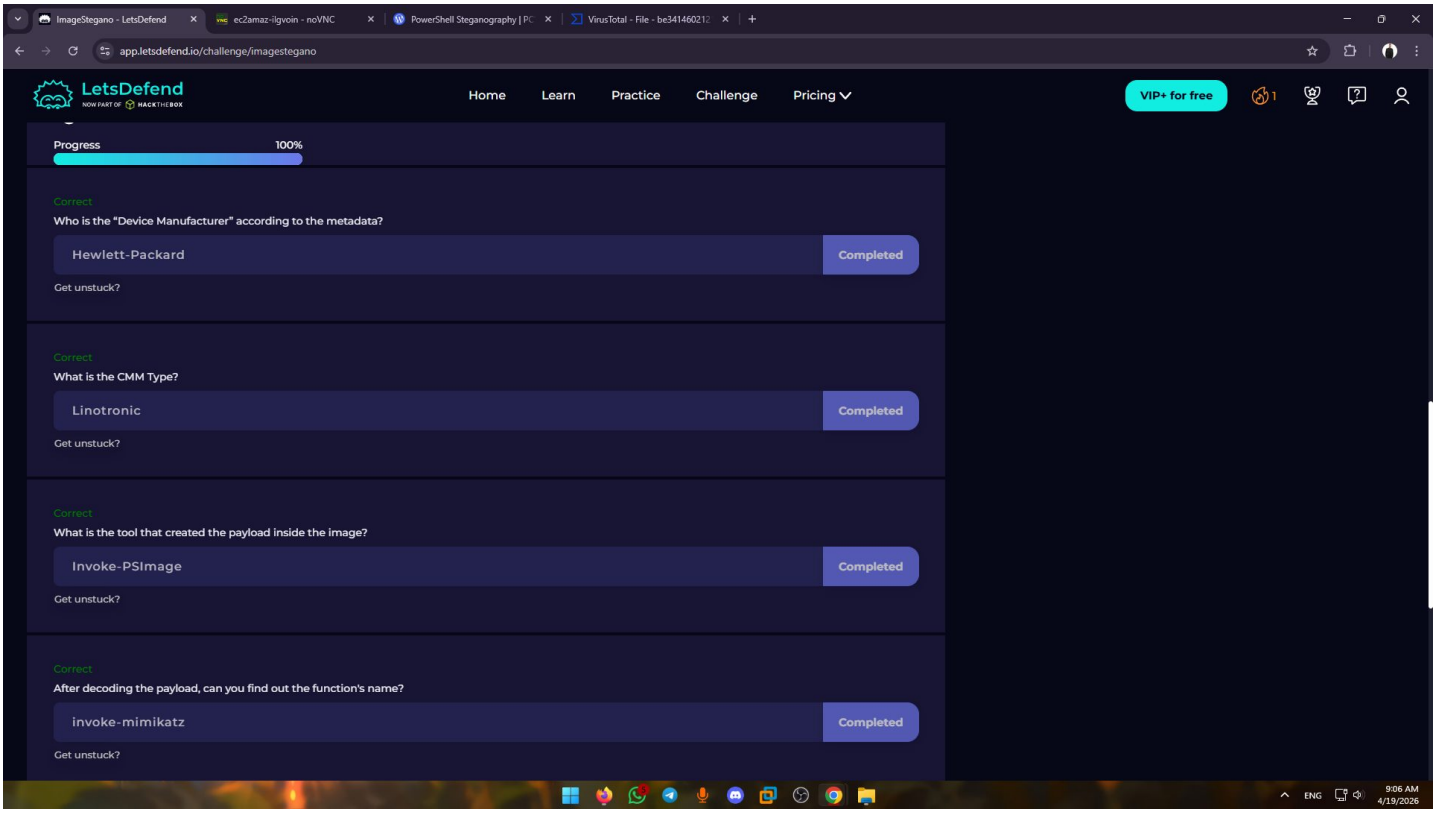

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Windows PowerShell
PSImage Decoder v1.0 [https://www.mertsarica.com]
[*] Decoding payload...
Settings
7:03 AM
4/19/2026
ENG
9:03 AM
4/19/2026

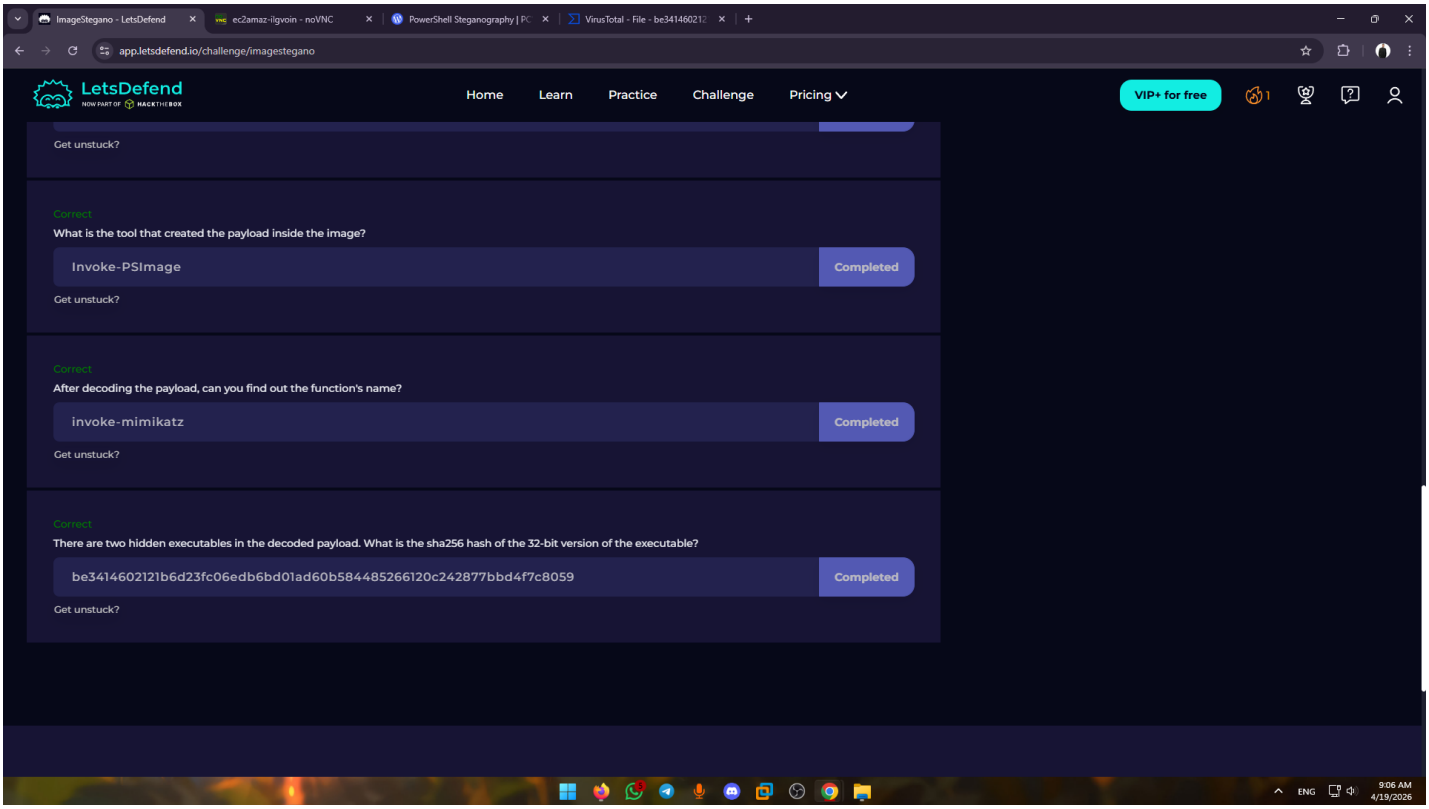

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.147.62.26&show_dot=true
Windows PowerShell
if ($ComputerName -eq $null -or $ComputerName -imatch "A\s*$")
Settings
7:04 AM
4/19/2026
ENG
9:03 AM
4/19/2026

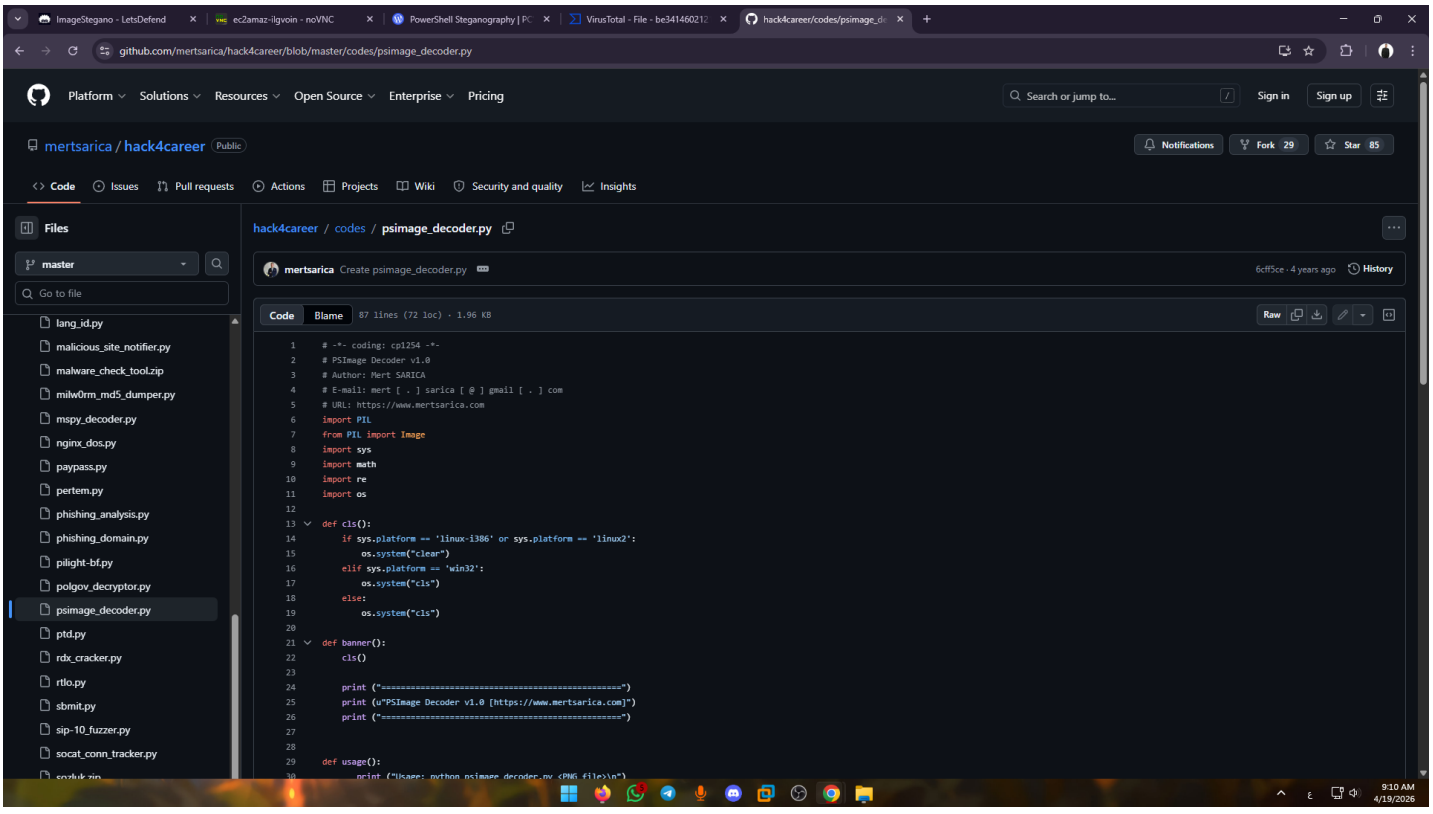

ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
VirusTotal - File - be3414602121
virustotal.com/gui/file/be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059/detection/f-be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059-1478821027
be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059
Sign in
Sign up
58
/ 72
Community Score
16
58/72 security vendors flagged this file as malicious
Reanalyze
Similar
More
be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059
decoded file32.exe.malz
pedll
checks-user-input
detect-debug-environment
Size
692.00 KB
Last Analysis Date
10 months ago
DLL
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY
13+
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Popular threat label
hacktool.mimikatz/mikatz
Threat categories
hacktool
trojan
pua
Family labels
mimikatz
mikatz
smgd
Security vendors' analysis
Do you want to automate checks?
AhnLab-V3
Trojan/Win32.Mimikatz.R235709
Alibaba
TrojanPSW:Win64/Mimikatz.9846a84e
AliCloud
HackTool:Win/mimikatz
ALYac
Misc.HackTool.Mimikatz
Arcabit
Trojan.Generic.D1DE2E20
Arctic Wolf
Unsafe
Avast
Win32:Malware-gen
AVG
Win32:Malware-gen
Avira (no cloud)
HEUR/AGEN.1302605
BitDefender
Trojan.GenericKD.31338016
Bkav Pro
W32.AIDetectMalware
ClamAV
Win.Trojan.Mimikatz-6466236-2
CrowdStrike Falcon
Win/malicious_confidence_100% (W)
CTX
Dll.hacktool.mimikatz
Cynet
Malicious (score: 100)
DeepInstinct
MALICIOUS
DrWeb
Tool.Mimikatz.159
Elastic
Windows.Hacktool.Mimikatz
ENG
9:04 AM
4/19/2026


ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
VirusTotal - File - be3414602121
virustotal.com/gui/file/be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059/details
be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059
Sign in
Sign up
NETAPI32.dll
ole32.dll
OLEAUT32.dll
RPCRT4.dll
SHLWAPI.dll
SAMLIB.dll
Secur32.dll
Exports
powershell_reflective_mimikatz
Contained Resources By Type
RT_MANIFEST 1
Contained Resources By Language
ENGLISH US 1
Contained Resources
SHA-256
File Type
Type
Language
Entropy
Chi2
59ba97d56a01766792386c3b379946bb613c8921e3daf8a878855a268ad5e4aa
unknown
RT_MANIFEST
ENGLISH US
5.04
5126.04
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How it Works
ToS | Privacy Notice
Blog | Releases
Community
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Contributors
Top Users
Community Buzz
Tools
API Scripts
YARA
Desktop Apps
Browser Extensions
Mobile App
Premium Services
Get a demo
Intelligence
Hunting
Graph
API v3 | v2
Documentation
Searching
Reports
API v3 | v2
Use Cases
ENG
9:05 AM
4/19/2026


ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
VirusTotal - File - be341460212
app.letsdefend.io/challenge/imagestegano
ImageStegano
You've gained a new badge!
Share with
Show Me!
ENG
9:06 AM
4/19/2026


ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
VirusTotal - File - be341460212
app.letsdefend.io/challenge/imagestegano
LetsDefend
NOW PART OF HACKTHEBOX
Home
Learn
Practice
Challenge
Pricing
VIP+ for free
Progress
100%
Correct
Who is the "Device Manufacturer" according to the metadata?
Hewlett-Packard
Completed
Get unstuck?
Correct
What is the CMM Type?
Linotronic
Completed
Get unstuck?
Correct
What is the tool that created the payload inside the image?
Invoke-PSImage
Completed
Get unstuck?
Correct
After decoding the payload, can you find out the function's name?
invoke-mimikatz
Completed
Get unstuck?
ENG
9:06 AM
4/19/2026


ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
VirusTotal - File - be34146021
app.letsdefend.io/challenge/imagestegano
LetsDefend
NOW PART OF HACKTHEBOX
Home
Learn
Practice
Challenge
Pricing
VIP+ for free
Get unstuck?
Correct
What is the tool that created the payload inside the image?
Invoke-PSImage
Completed
Get unstuck?
Correct
After decoding the payload, can you find out the function's name?
invoke-mimikatz
Completed
Get unstuck?
Correct
There are two hidden executables in the decoded payload. What is the sha256 hash of the 32-bit version of the executable?
be3414602121b6d23fc06edb6bd01ad60b584485266120c242877bbd4f7c8059
Completed
Get unstuck?
ENG
9:06 AM
4/19/2026


ImageStegano - LetsDefend
ec2amaz-ilgvoin - noVNC
PowerShell Steganography | PC
VirusTotal - File - be34146021
hack4career/codes/psimage_de
github.com/mertsarica/hack4career/blob/master/codes/psimage_decoder.py
Platform
Solutions
Resources
Open Source
Enterprise
Pricing
Search or jump to...
Sign in
Sign up
mertsarica / hack4career
Public
Notifications
Fork 29
Star 85
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Files
master
Go to file
lang_id.py
malicious_site_notifier.py
malware_check_tool.zip
milw0rm_md5_dumper.py
mspy_decoder.py
nginx_dos.py
paypass.py
pertem.py
phishing_analysis.py
phishing_domain.py
pilight-bf.py
polgov_decryptor.py
psimage_decoder.py
ptd.py
rdx_cracker.py
rtlo.py
sbmit.py
sip-10_fuzzer.py
socat_conn_tracker.py
hack4career / codes / psimage_decoder.py
mertsarica Create psimage_decoder.py
6cff5ce · 4 years ago
History
Code
Blame
87 lines (72 loc) · 1.96 KB
Raw
-*- coding: cp1254 -*-
PSImage Decoder v1.0
Author: Mert SARICA
E-mail: mert [.] sarica [@] gmail [.] com
URL: https://www.mertsarica.com
import PIL
from PIL import Image
import sys
import math
import re
import os
def cls():
if sys.platform == 'linux-i386' or sys.platform == 'linux2':
os.system("clear")
elif sys.platform == 'win32':
os.system("cls")
else:
os.system("cls")
def banner():
cls()
print ("==")
print (u"PSImage Decoder v1.0 [https://www.mertsarica.com]")
print ("==")
def usage():
print ("Usage: python psimage_decoder.py <PNG file>\n")
9:10 AM
4/19/2026